Unit 3 - Calculus
3.1 Limits and Derivatives

Graham West

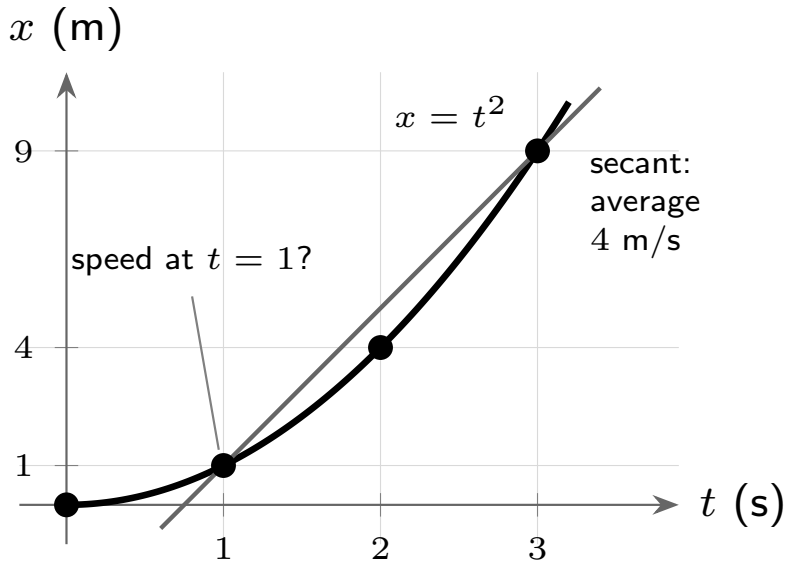
MSDS 520: Math and Stats Foundations for Data Science
Meharry Medical College

Outline

1. Calculus: the Study of Change
2. Average Rate of Change
3. From Secant to Tangent
4. The Derivative
5. Derivative Rules
6. The Chain Rule
7. Continuity and Differentiability
8. The Second Derivative
9. Partial Derivatives
10. The Gradient
11. Review

Calculus: the Study of Change

A Problem to Motivate Us



A car pulls away from a stop sign

Position $x(t) = t^2$ m after t s. From $t = 1$ to $t = 3$ it covers 8 m in 2 s: an **average** of 4 m/s.

But a speedometer reads one instant

What does it read at exactly $t = 1$? Speed needs **two** instants; at one, distance and time are both 0:

$$\text{speed} = \frac{0}{0} \quad ???$$

This lecture's question

How do we measure a rate of change at a **single instant**? $\Rightarrow$ the **derivative**

Where This Shows Up in Data Science

Rates of change are everywhere

- **Velocity**: meters per second.
- **Drug elimination**: how fast a drug's blood concentration falls, mg/L per hour.
- **Epidemics**: new cases per day, and whether that number is itself growing.
- **Temperature across a room**: degrees per meter. A rate can be with respect to position, not just time.

Why a data scientist needs it

- Training a model means asking: if I nudge this **weight**, how fast does the **error** change?
- That rate is a derivative. Every model you train, from linear regression to a neural network, is trained by following derivatives downhill (Unit 3-2).

What We Will Build

The plan

1. **Average** rate of change: secant lines and the **difference quotient**.
2. Shrink the interval: **just enough** about limits to get past $0/0$.
3. The **derivative**: definition, notation, and six worked examples.
4. **Rules** (power, sum, product, quotient): shortcuts, so we never have to take the limit by hand again.
5. The **chain rule**, for functions inside functions. Every error function in Unit 3-2 needs it.
6. **Continuity and differentiability**: where a derivative fails, x^2 versus $|x|$.
7. The **second derivative**: curvature, convex and concave, inflection points.
8. **Partial derivatives** and the **gradient**: a slope in each direction for $f(x, y)$ (Unit 1-3), stacked into a vector that points uphill.

Notebooks for this lecture: `U3-2_Limits-1` and `-2`, `U3-3_Derivatives-1` to `-3`, `U3-6_Multivariable-1` to `-3`.

Average Rate of Change

Rate of Change Is Rise over Run

Definition: average rate of change

The **average rate of change** of f from x_1 to x_2 is

$$\frac{\Delta y}{\Delta x} = \frac{f(x_2) - f(x_1)}{x_2 - x_1} = \frac{\text{rise}}{\text{run}}.$$

- Δ ("delta") means "change in".
- Units: output units **per** input unit.

The car

$$\frac{x(3) - x(1)}{3 - 1} = \frac{9 - 1}{2} = 4 \text{ m/s}$$

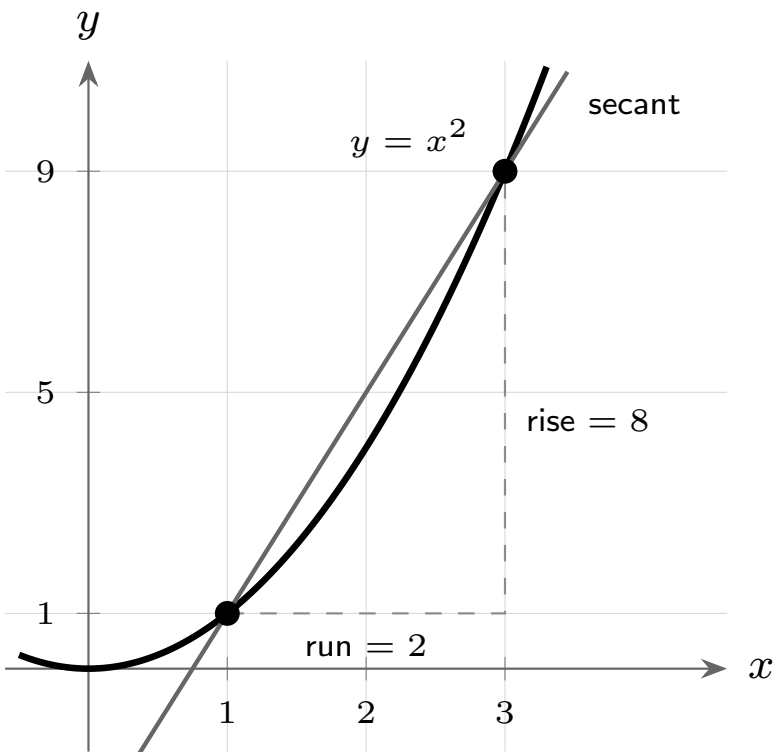
A drug clearing the blood

C falls from 8 to 2 mg/L over 3 hours:

$$\frac{2 - 8}{3} = -2 \text{ mg/L per hour}$$

Negative: the concentration is **falling**.

Secant Lines



Definition: secant line

A **secant line** crosses a graph at **two** points.

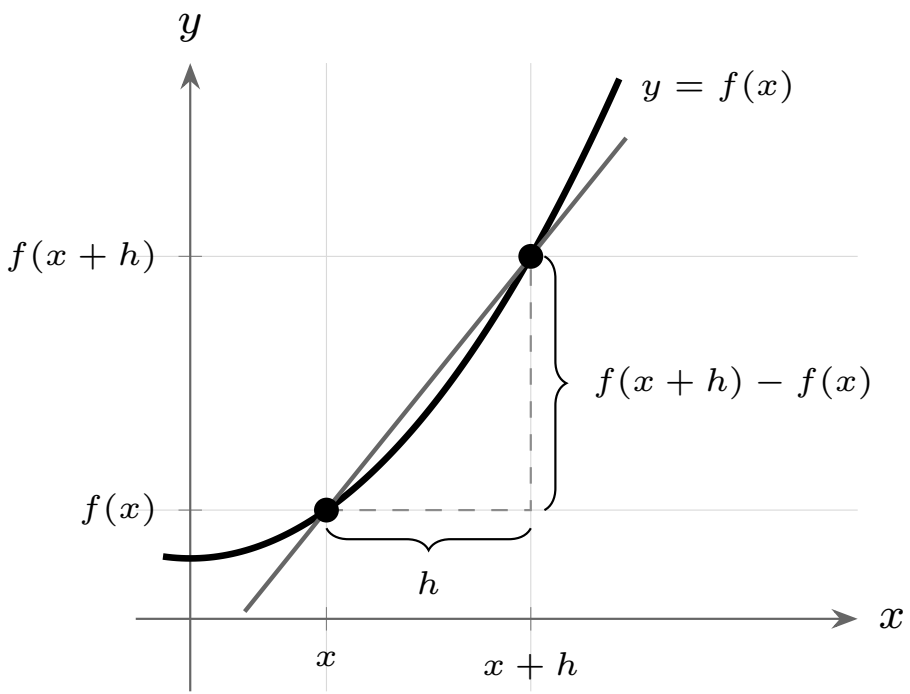
- Its slope is the **average** rate of change between them.
- Here: from $(1, 1)$ to $(3, 9)$,

$$\text{slope} = \frac{8}{2} = 4.$$

The catch

The curve is steeper at $x = 3$ than at $x = 1$. A single average hides that.

The Difference Quotient



Same slope formula, new names: start at x, take a step of size h.

Definition: difference quotient

$$\frac{f(x+h) - f(x)}{h}$$

- h is the **run**: the step.
- $f(x+h) - f(x)$ is the **rise**.
- It is the slope of the secant from x to $x + h$.

Where this is going

Make h smaller and smaller.

Exercise: Average Rates of Change

Try it

1. For $f(x) = x^2$, compute the difference quotient at $x = 1$ with $h = 1$, then with $h = 0.5$. What is happening to the slope?
2. A patient's temperature is $101.3°$F at 8 am and $99.5°$F at 2 pm. Find the average rate of change, with units, and say what it means.
3. For the line $f(x) = 3x + 2$, simplify $\dfrac{f(x+h) - f(x)}{h}$ for **any** h. What do you notice?

Exercise: Solution

Answers

1. With $h = 1$: $\dfrac{f(2) - f(1)}{1} = \dfrac{4 - 1}{1} = 3.$

   With $h = 0.5$: $\dfrac{f(1.5) - f(1)}{0.5} = \dfrac{2.25 - 1}{0.5} = 2.5.$

   As h shrinks, the secant gets **less steep**: it is settling toward the steepness right at $x = 1$.

2. Six hours pass, so

   $$\frac{99.5 - 101.3}{6} = \frac{-1.8}{6} = -0.3 \ ^\circ\text{F per hour.}$$

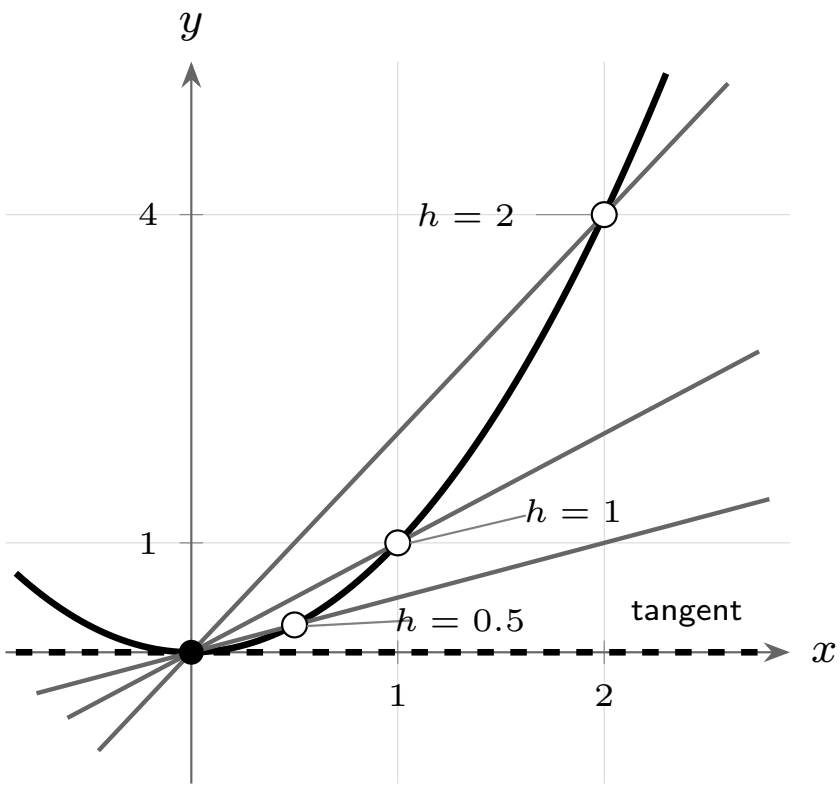
   The fever fell, on average, 0.3 degrees every hour.

3. $\dfrac{3(x+h) + 2 - (3x+2)}{h} = \dfrac{3x + 3h + 2 - 3x - 2}{h} = \dfrac{3h}{h} = 3.$

   The answer is 3 for every h. A secant of a straight line is the line itself, so its slope never changes.

From Secant to Tangent

Shrink the Step



$f(x) = x^2$ at $x = 0$

h	secant slope
2	2
1	1
0.5	0.5
0.1	0.1
0.01	0.01

What we see

- The points labelled by h slide down the curve toward $(0, 0)$.
- The secants flatten onto the dashed tangent, the x-axis: their slopes home in on 0.

Notebook `U3-3_Derivatives-1_SecantToTangent`, §2 and §3

The Problem with $h = 0$

Why not just set $h = 0$?

$$\frac{(0+0)^2 - 0^2}{0} = \frac{0}{0}$$

- Zero rise over zero run: the speedometer problem again.
- $0/0$ is **undefined**. It is not 0, not 1, not anything.

The way around it

- We may **not** set $h = 0$.
- We **may** make h as close to 0 as we like, and watch where the slope is heading.
- "Where it is heading" has a name: a **limit**.

Limits: Just Enough

Definition (informal): limit

$$\lim_{h \to 0} g(h) = L$$

means $g(h)$ gets **as close to** L **as we like** once h is close enough to 0, **without** h **ever equalling** 0. Read it "the limit of $g(h)$ as h goes to 0".

From both sides

h	slope	h	slope
0.1	0.1	-0.1	-0.1
0.01	0.01	-0.01	-0.01
0.001	0.001	-0.001	-0.001

Both sides head to the same place: the limit is 0.

What a limit ignores

- The value **at** $h = 0$ itself.
- That is the whole point: the quotient **has** no value there, yet it still heads somewhere.

Notebook `U3-2_Limits-1_ApproachingALimit`, §1 and §3

Computing a Limit: Simplify, Then Shrink

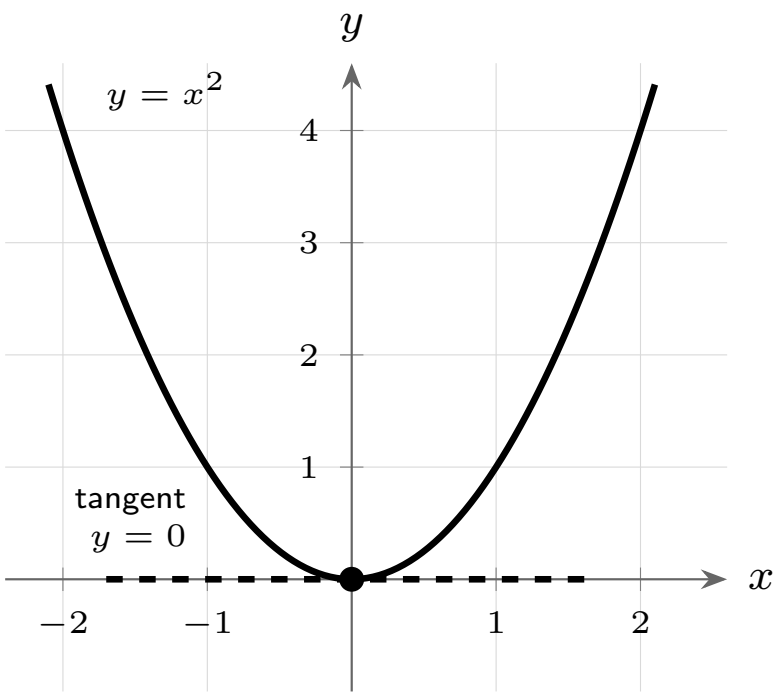
Worked: the slope of x^2 at $x = 0$

$$\begin{aligned}
\frac{(0+h)^2 - 0^2}{h} &= \frac{h^2}{h} && \text{expand} \\
&= \frac{h \cdot h}{h} && \text{factor out } h \\
&= h && \text{cancel, allowed since } h \neq 0 \\
\lim_{h \to 0} h &= 0 && \text{now let } h \text{ shrink}
\end{aligned}$$

The move

- The h that caused $0/0$ **cancels**. After that, nothing stops h from going to 0.
- Every limit in this unit goes this way: simplify first, then let h shrink.
- The same steps work at any point. The next section runs them for every x at once.

The Tangent Line



Definition: tangent line

The line the secants approach as $h \to 0$.

- Touches the graph at **one** point.
- Runs in the **same direction** as the curve.
- Slope $=$ the **instantaneous** rate of change.

Back to the car

At $t = 0$ the tangent is flat: the speedometer reads **0 m/s**, the car at rest as it pulls away. The $t = 1$ reading comes in the next section.

Notebook `U3-3_Derivatives-1_SecantToTangent`, §4

The Derivative

The Derivative

Definition: derivative

The **derivative** of f at x is the limit of the difference quotient:

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

- The slope of the **tangent line** at x.
- The **instantaneous rate of change** of f at x, in output units **per** input unit: for a drug level over time, mg/L per hour.
- To **differentiate** is to find the derivative. If the limit exists, f is **differentiable** at x.

Secant versus tangent

	secant	**tangent**
points on the graph	two	one
step	$h \neq 0$	$h \to 0$
slope means	average rate	instantaneous rate

Notation

Four ways to write the same derivative

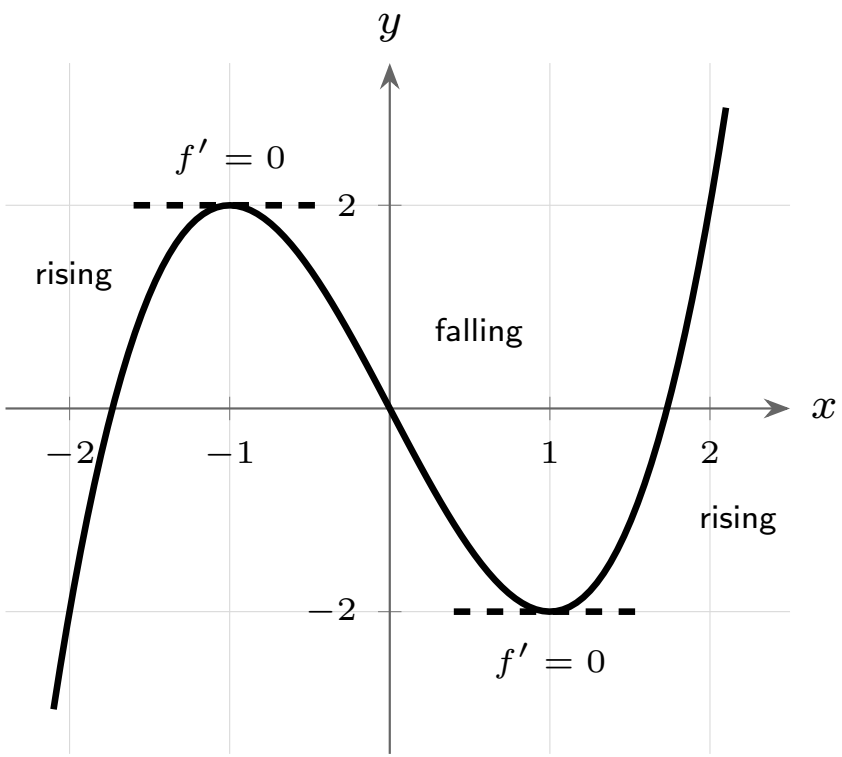
$f'(x)$	"f prime of x"
$\dfrac{df}{dx}$	a tiny rise df over a tiny run dx: the limit of $\Delta f / \Delta x$
$\dfrac{d}{dx} f(x)$	an operation applied to f: in goes a function, out comes a function
$\dot{x}(t)$	"x dot": only for rates **with respect to time**, $\dot{x} = \dfrac{dx}{dt}$

If $y = f(x)$, you will also see y' and $\dfrac{dy}{dx}$. The car's velocity is $\dot{x}(t)$.

Several inputs: partial derivatives

$\dfrac{\partial f}{\partial x}$, "partial f, partial x". The curly ∂ says f has **other** inputs, held fixed while only x moves. The partial derivatives section, later in this lecture, builds this.

Reading a Derivative off a Graph



The sign of f'

$f'(x) > 0$ graph **rising**

$f'(x) < 0$ graph **falling**

$f'(x) = 0$ graph **flat**

Why we care

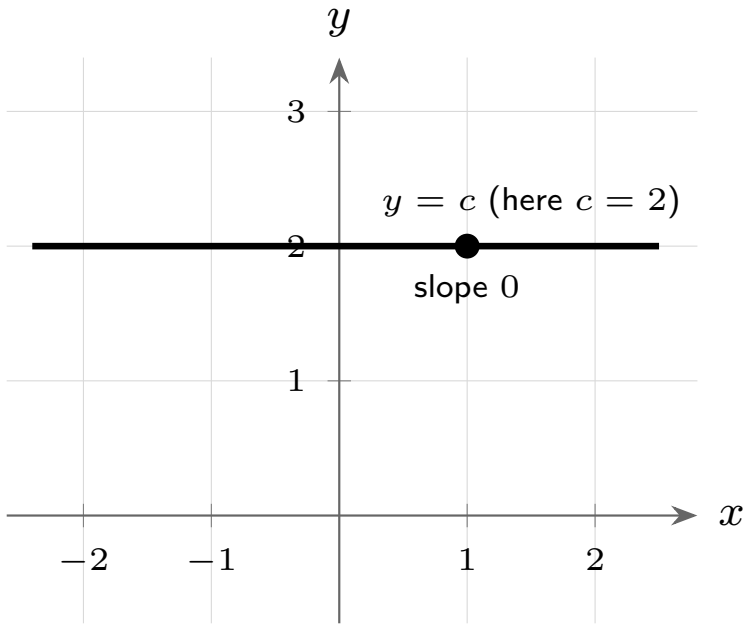
The flat spots are the peaks and valleys. Finding the lowest valley is how models are fit: Unit 3-2 hunts for these.

Example: $f(x) = c$

From the definition

$f(x+h) = c$ too: there is no x to replace.

$$f'(x) = \lim_{h \to 0} \frac{c - c}{h}$$
$$= \lim_{h \to 0} \frac{0}{h} = 0$$



What it says

A constant never changes, so its rate of change is 0 everywhere. The graph is flat, and the tangent at every point is the line itself.

Watch out

$\frac{0}{h} = 0$ because $h \neq 0$. The limit never divides by zero.

Example: $f(x) = x$

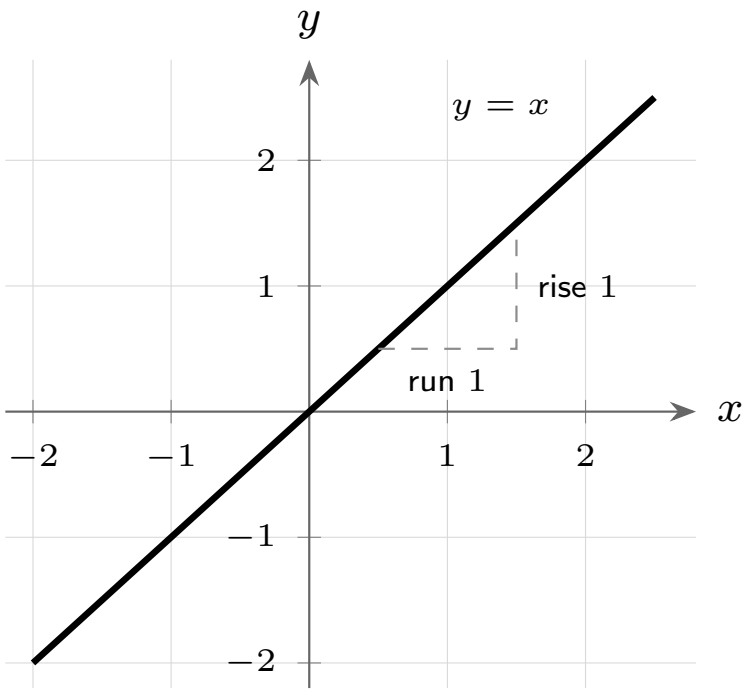
From the definition

$f(x+h) = x + h.$

$$f'(x) = \lim_{h \to 0} \frac{(x+h) - x}{h}$$
$$= \lim_{h \to 0} \frac{h}{h} = 1$$

What it says

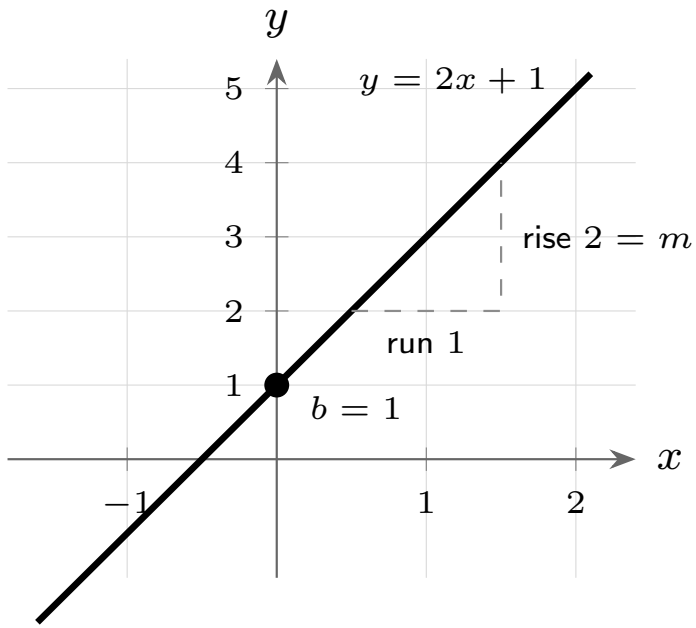
The line $y = x$ has slope 1 everywhere: up one for every one across.



Example: $f(x) = mx + b$

From the definition

$$
\begin{aligned}
f'(x) &= \lim_{h \to 0} \frac{m(x+h) + b - (mx + b)}{h} \\
&= \lim_{h \to 0} \frac{mx + mh + b - mx - b}{h} \\
&= \lim_{h \to 0} \frac{mh}{h} = m
\end{aligned}
$$



What it says

- A line's derivative is its **slope** m, the same at every x. The intercept b slides the line up or down without tilting it, so it drops out.
- c and x were the special cases $m = 0$, and $m = 1$ with $b = 0$.

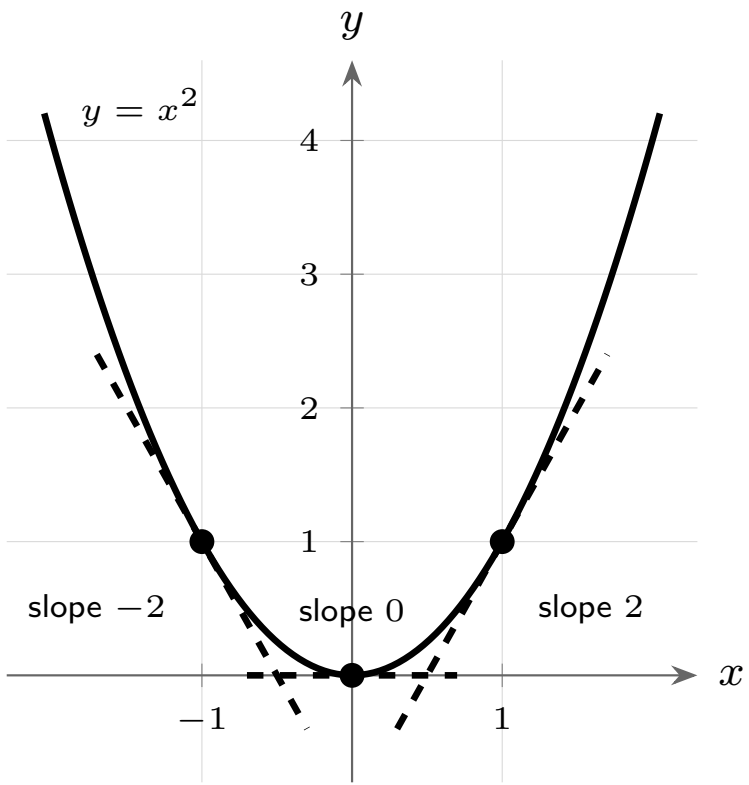
Example: $f(x) = x^2$

From the definition

$$\begin{aligned} f'(x) &= \lim_{h \to 0} \frac{(x+h)^2 - x^2}{h} \\ &= \lim_{h \to 0} \frac{x^2 + 2xh + h^2 - x^2}{h} \\ &= \lim_{h \to 0} \frac{h\,(2x+h)}{h} \\ &= \lim_{h \to 0} (2x + h) = 2x \end{aligned}$$

The derivative is a new function

$f'(x) = 2x$ gives the slope at **every** x at once: 0 at $x = 0$, then $2, 4, 6$ at $x = 1, 2, 3$. The car's speedometer at $t = 1$: **2 m/s**.



Example: $f(x) = \sin x$

From the definition

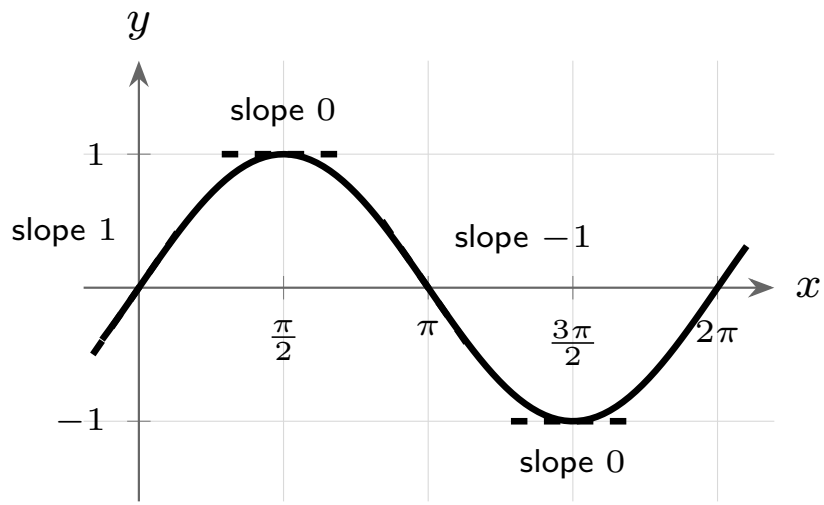
The angle-addition identity, $\sin(x+h) = \sin x \cos h + \cos x \sin h$, splits the quotient:

$$\frac{\sin(x+h) - \sin x}{h} = \sin x \cdot \frac{\cos h - 1}{h} + \cos x \cdot \frac{\sin h}{h}$$

Two limits, read off a table:

h	$\frac{\sin h}{h}$	$\frac{\cos h - 1}{h}$
0.1	0.9983	-0.0500
0.01	0.99998	-0.0050
$\to 0$	$\to 1$	$\to 0$

So $f'(x) = \sin x \cdot 0 + \cos x \cdot 1 = \cos x$.



What it says

The slopes $1, 0, -1, 0$ are exactly the values of $\cos x$ at $0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$. (This needs x in radians.)

Example: $f(x) = 1/x$

From the definition

Combine the fractions in the numerator first:

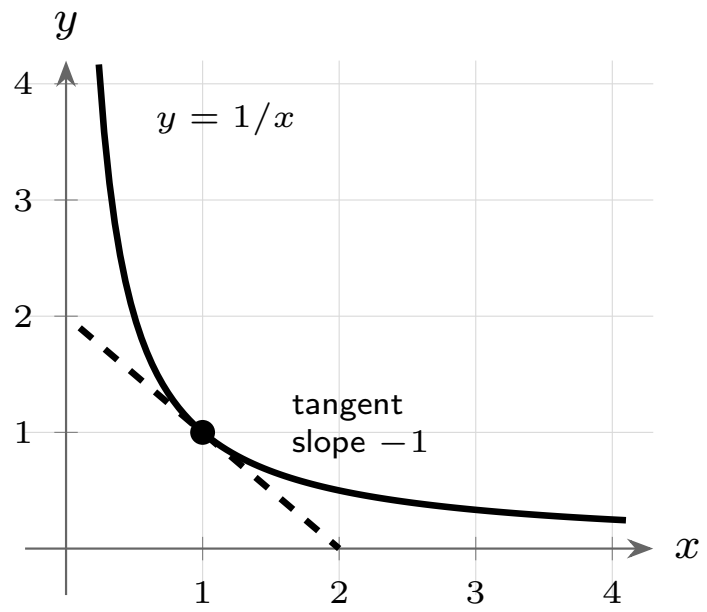
$$\frac{1}{x+h} - \frac{1}{x} = \frac{x-(x+h)}{x\,(x+h)} = \frac{-h}{x\,(x+h)}$$

Now divide by h, which cancels:

$$\frac{f(x+h)-f(x)}{h} = \frac{-1}{x\,(x+h)}$$

Let $h \to 0$:

$$f'(x) = -\frac{1}{x^2}$$



What it says

- $-1/x^2 < 0$: $1/x$ is **always falling**.
- Steep near 0, nearly flat far out.

Exercise: From the Definition

Try it

1. Use the limit definition to find the derivative of $f(x) = 3x^2$.
2. Use the limit definition to find the derivative of $f(x) = x^2 + x$.
3. A drug's blood concentration has $\dfrac{dC}{dt} = -1.5$ mg/L per hour at $t = 4$ hours. In words, what does that mean?

Exercise: Solution

Answers

1. $$\frac{3(x+h)^2 - 3x^2}{h} = \frac{3x^2 + 6xh + 3h^2 - 3x^2}{h} = \frac{h\,(6x + 3h)}{h}$$
   $= 6x + 3h \;\to\; 6x.$
   That is 3 times the derivative of x^2: the constant just rides along.

2. $$\frac{(x+h)^2 + (x+h) - x^2 - x}{h} = \frac{2xh + h^2 + h}{h} = 2x + h + 1 \;\to\; 2x + 1.$$
   That is the derivative of x^2 plus the derivative of x: the sum splits.

3. At 4 hours the concentration is **falling**, at 1.5 mg/L per hour. If that pace held for one more hour, it would drop by about 1.5 mg/L.

Hold on to answers 1 and 2

They are two of the rules in the next section.

Derivative Rules

The Power Rule

Power rule

$$\frac{d}{dx}\,x^n = n\,x^{n-1} \qquad \text{for any number } n$$

Examples

function		as a power		derivative		simplified
x^3			$\longrightarrow$	$3x^2$		
x^{10}			$\longrightarrow$	$10x^9$		
$\sqrt{x}$	$=$	$x^{1/2}$	$\longrightarrow$	$\frac{1}{2}x^{-1/2}$	$=$	$\dfrac{1}{2\sqrt{x}}$
$\dfrac{1}{x^2}$	$=$	x^{-2}	$\longrightarrow$	$-2x^{-3}$	$=$	$-\dfrac{2}{x^3}$

Watch out

Rewrite roots and fractions as powers **first**: a root is a fractional power, a fraction a negative power.

Constant Multiple and Sum Rules

Constant multiple rule

$$\frac{d}{dx}\big[c\,f(x)\big] = c\,f'(x)$$

Constants ride along.

Sum rule

$$\frac{d}{dx}\big[f(x) \pm g(x)\big] = f'(x) \pm g'(x)$$

Sums and differences split.

Worked: a polynomial, term by term

$$\begin{aligned}\frac{d}{dx}\left(4x^3 - 2x + 7\right) &= 4 \cdot 3x^2 \;-\; 2 \cdot 1 \;+\; 0 \\ &= 12x^2 - 2\end{aligned}$$

Power rule on each term, constants carried, and the lone 7 (a constant) differentiates to 0.

Common Derivatives

Table

$f(x)$	$f'(x)$
c	0
x^n	$n\,x^{n-1}$
e^x	e^x
$\ln x$	$\dfrac{1}{x} \quad (x > 0)$
a^x	$a^x \ln a$
$\log_a x$	$\dfrac{1}{x \ln a} \quad (x > 0)$
$\sin x$	$\cos x$
$\cos x$	$-\sin x$

The Product Rule

Product rule

$$\frac{d}{dx}\big[f(x)\,g(x)\big] = f'(x)\,g(x) + f(x)\,g'(x)$$

In words: (first)′ × second + first × (second)′.

Worked: $x^2 e^x$

	$f = x^2, \quad g = e^x$	name the first and the second
	$f' = 2x, \quad g' = e^x$	differentiate each one
$\frac{d}{dx}\big[x^2 e^x\big]$	$= 2x \cdot e^x + x^2 \cdot e^x$	assemble $f'g + fg'$
	$= x\,e^x\,(2 + x)$	factor out $x\,e^x$

Notebook `U3-3_Derivatives-2_Rules`, §3

The Quotient Rule

Quotient rule

$$\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{f'(x)\,g(x) - f(x)\,g'(x)}{g(x)^2}$$

In words: "low d-high minus high d-low, over low squared" (low = bottom, high = top, d = derivative of).

Worked: $x/(x+1)$

$$\frac{d}{dx}\left[\frac{x}{x+1}\right]$$

$f = x, \quad g = x+1$ — name the top and the bottom

$f' = 1, \quad g' = 1$ — differentiate each one

$= \dfrac{1\cdot(x+1) - x\cdot 1}{(x+1)^2}$ — assemble $\dfrac{f'g - fg'}{g^2}$

$= \dfrac{x+1-x}{(x+1)^2} = \dfrac{1}{(x+1)^2}$ — simplify the top

Exercise: Rules

Try it: differentiate

1. $5x^4 - 3x^2 + 8$
2. $2\sqrt{x} + \dfrac{1}{x}$
3. $3e^x - 4\ln x$
4. $x^3 \ln x$
5. $\dfrac{e^x}{x}$
6. $(x+1)^2$ (expand it first)

Exercise: Solution

Answers

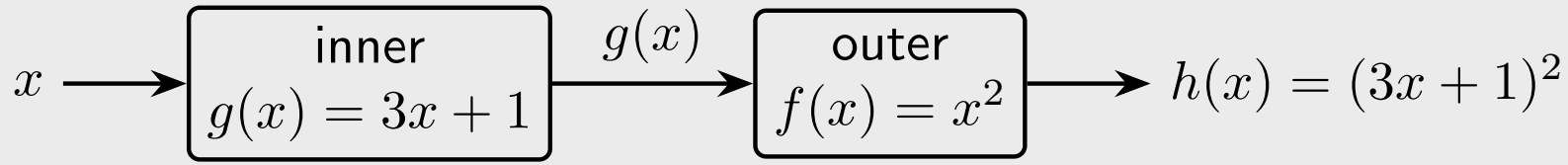
1. Term by term: $5 \cdot 4x^3 - 3 \cdot 2x + 0 = 20x^3 - 6x$.
2. As powers, $2x^{1/2} + x^{-1}$, so $2 \cdot \frac{1}{2}x^{-1/2} - x^{-2} = \dfrac{1}{\sqrt{x}} - \dfrac{1}{x^2}$.
3. From the table: $3e^x - 4 \cdot \dfrac{1}{x} = 3e^x - \dfrac{4}{x}$.
4. Product, with $f = x^3$, $g = \ln x$: $\quad 3x^2 \ln x + x^3 \cdot \dfrac{1}{x} = 3x^2 \ln x + x^2$.
5. Quotient, with top e^x, bottom x: $\quad \dfrac{e^x \cdot x - e^x \cdot 1}{x^2} = \dfrac{e^x(x-1)}{x^2}$.
6. $(x+1)^2 = x^2 + 2x + 1$, so the derivative is $2x + 2$. Expanding would not work for $(x+1)^{50}$: the chain rule, next, does it without expanding.

The Chain Rule

Functions Inside Functions

Composition, from Unit 1-3

$h(x) = (f \circ g)(x) = f\big(g(x)\big)$: the **inner** function g acts first, and the **outer** function f acts on its output.



Reading a composition backward

function h	inner g	outer f
$e^{-0.2t}$	$-0.2t$	e^t
$\sqrt{x^2 + 1}$	$x^2 + 1$	$\sqrt{x}$
$(5 - 2w)^2$	$5 - 2w$	w^2

Why we care

A model's error is a square of (actual minus predicted): the last row. Unit 3-2 differentiates exactly this.

The Chain Rule

Chain rule

$$h(x) = f\big(g(x)\big) \quad \Longrightarrow \quad h'(x) = f'\big(g(x)\big) \cdot g'(x)$$

In words: the derivative of the outer, at the inner, times the derivative of the inner.

Worked: $h(x) = (3x+1)^2$

	$g(x) = 3x+1, \quad f(x) = x^2$	name the inner and the outer
	$g'(x) = 3, \quad f'(x) = 2x$	differentiate each one
	$f'\big(g(x)\big) = 2(3x+1)$	outer's derivative, at the inner
$h'(x)$	$= 2(3x+1) \cdot 3 = 18x+6$	times $g'(x)$

Notebook `U3-3_Derivatives-2_Rules`, §4

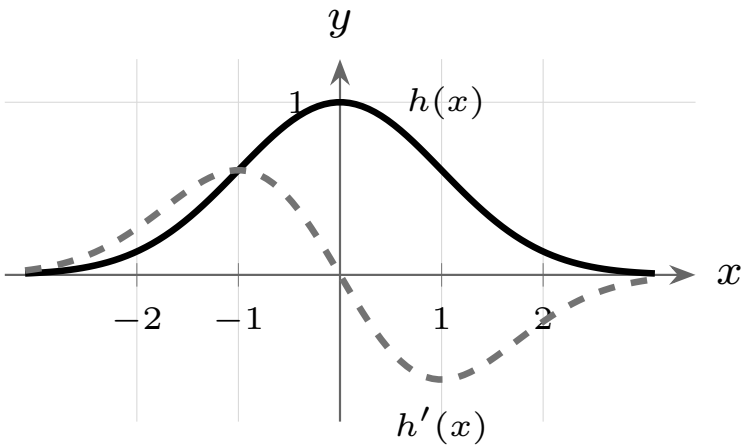
The Chain Rule: the Bell Curve

Worked: $h(x) = e^{-x^2/2}$, the shape of the normal distribution

$$g(x) = -\tfrac{1}{2}x^2, \quad f(x) = e^x$$ name the inner and the outer

$$g'(x) = -x, \quad f'(x) = e^x$$ differentiate each one

$$h'(x) = e^{-x^2/2} \cdot (-x) = -x\, e^{-x^2/2}$$ assemble $f'\big(g(x)\big) \cdot g'(x)$



What it says

- $h' > 0$ left of 0 (rising), $h' < 0$ right of 0 (falling), and $h'(0) = 0$ at the peak.
- The bell is steepest at $x = \pm 1$: one standard deviation from the center.

Exercise: Chain Rule

Try it: differentiate

1. $(2x - 5)^3$
2. e^{3x}
3. $\sqrt{x^2 + 1}$

Exercise: Solution

Answers

1. Inner $g = 2x - 5$ ($g' = 2$), outer $f = x^3$ ($f' = 3x^2$):
   $3(2x-5)^2 \cdot 2 = 6(2x-5)^2$.
2. Inner $g = 3x$ ($g' = 3$), outer $f = e^x$ ($f' = e^x$): $\quad e^{3x} \cdot 3 = 3e^{3x}$.
3. Inner $g = x^2 + 1$ ($g' = 2x$), outer $f = x^{1/2}$ ($f' = \frac{1}{2}x^{-1/2}$):

$$\tfrac{1}{2}(x^2+1)^{-1/2} \cdot 2x = \frac{x}{\sqrt{x^2+1}}.$$

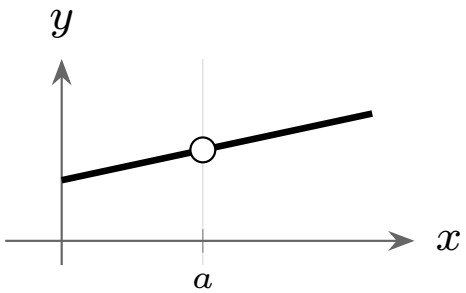
Continuity and Differentiability

Continuity: No Lifting the Pen

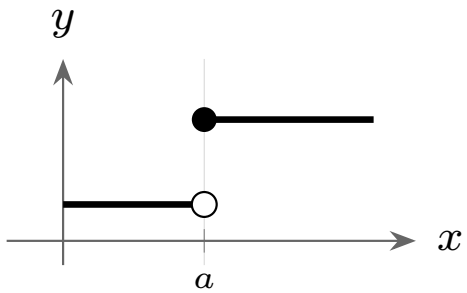
Definition: continuous at a point

f is **continuous** at $x = a$ when you can draw through it without lifting your pen. All three must hold:

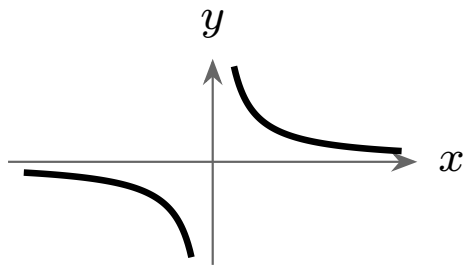
1. $f(a)$ exists.
2. $\lim_{x \to a} f(x)$ exists: x closes in on a, and both sides head to the same place.
3. The two are equal.



Hole: $f(a)$ is missing.



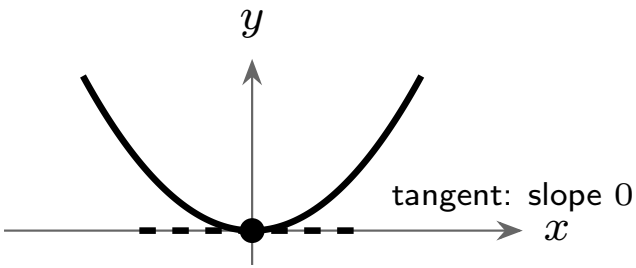
Jump: the sides disagree, so no limit.

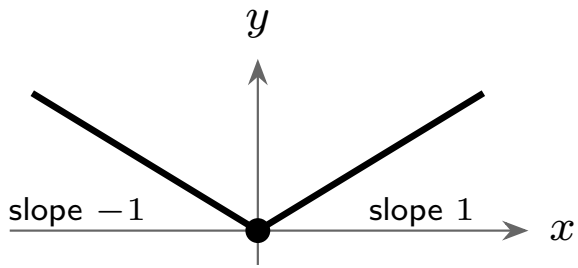


Blow-up: $1/x$ at 0 has no finite limit.

Notebook `U3-2_Limits-2_ContinuityAndDiscontinuity`, §1 and §2

x^2 Versus $|x|$: Is There a Slope at 0?





$f(x) = x^2$ at 0

$$\frac{(0+h)^2 - 0^2}{h} = \frac{h^2}{h} = h$$

h	slope	h	slope
0.1	0.1	-0.1	-0.1
0.01	0.01	-0.01	-0.01

Both sides head to 0: $f'(0) = 0$.

$f(x) = |x|$ at 0

$$\frac{|0+h| - |0|}{h} = \frac{|h|}{h}$$

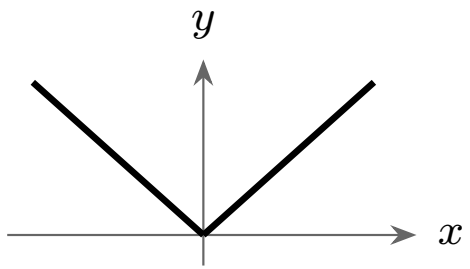
h	slope	h	slope
0.1	1	-0.1	-1
0.01	1	-0.01	-1

The sides disagree: **no limit**, no $f'(0)$.

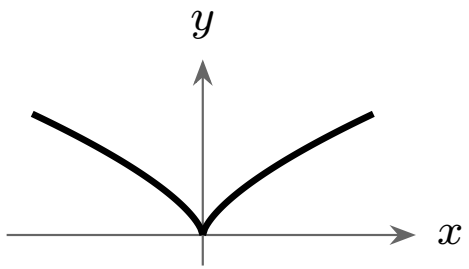
What it says

Both are continuous, and both bottom out at 0. Only x^2 is **differentiable** there.

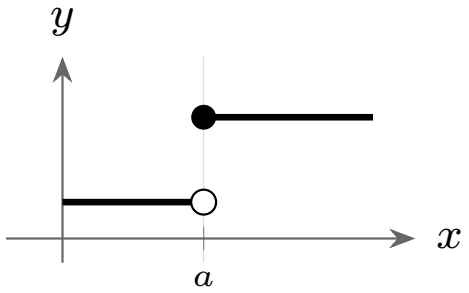
Where the Derivative Fails



Corner: $|x|$ at 0. The two sides have different slopes.



Cusp: $|x|^{2/3}$ at 0. Slopes shoot to $-\infty$ and $+\infty$.



Discontinuity: the rise stays fixed as the run shrinks to 0.

Continuity and differentiability: a one-way street

- **Differentiable** at $a \Rightarrow$ **continuous** at a: a break always kills the derivative.
- The reverse is **false**: $|x|$ is continuous at 0, yet has no derivative there.
- A curve with no corners, cusps, or breaks is called **smooth**.

Exercise: Continuity and Differentiability

Try it

1. For $f(x) = |x - 2|$, write the difference quotient at $x = 2$ and simplify. Is f continuous at 2? Differentiable at 2?
2. Repeat for $g(x) = (x - 2)^2$.
3. Both f and g have their lowest point at $x = 2$. Which one can you find by setting the derivative to 0?
4. The step function is $H(x) = 0$ for $x < 0$ and $H(x) = 1$ for $x \geq 0$. Is it continuous at 0? Can it be differentiable there?

Exercise: Solution

Answers

1. $\dfrac{|2+h-2|-|2-2|}{h} = \dfrac{|h|}{h}$, which is 1 for $h > 0$ and -1 for $h < 0$.

   The sides disagree, so there is no limit: **not differentiable** at 2. Yet $f(2) = 0$ and both sides head to 0, so it is **continuous**. It is a corner.

2. $\dfrac{(2+h-2)^2-(2-2)^2}{h} = \dfrac{h^2}{h} = h \ \to \ 0.$

   So $g'(2) = 0$: **differentiable**, and therefore **continuous**.

3. Only g: $g'(2) = 0$ marks the bottom. $f'(2)$ does not exist, so there is nothing to set to 0.

4. From the left H heads to 0, from the right to 1: no limit, so **not continuous**. Differentiable implies continuous, so it **cannot** be differentiable at 0.

The Second Derivative

The Derivative of the Derivative

Definition: second derivative

$f'(x)$ is itself a function, so it has a derivative too: the **second derivative**.

$$f''(x) = \frac{d}{dx} f'(x)$$

- It measures how fast the **slope itself** is changing.
- Units: output units per input unit, **per input unit again**.

Notation

$f''(x)$	"f double prime of x"
$\frac{d^2 f}{dx^2}$	short for $\frac{d}{dx}\left(\frac{df}{dx}\right)$
$\ddot{x}(t)$	"x double dot": for time only

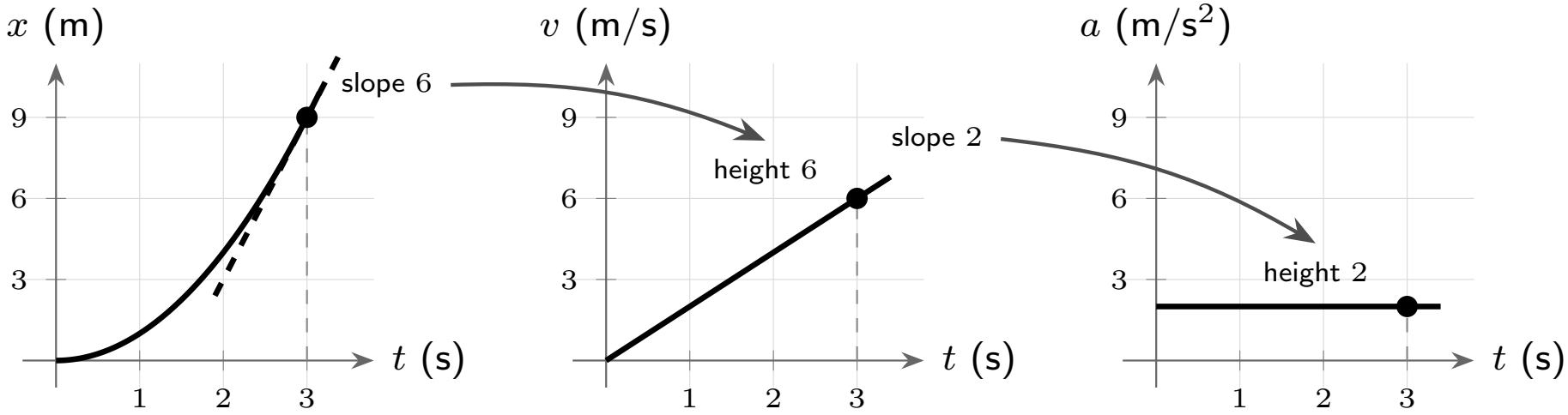
Differentiate twice

$$f(x) = x^3 - 3x$$
$$f'(x) = 3x^2 - 3$$
$$f''(x) = 6x$$

Notebook `U3-3_Derivatives-3_SecondDerivativeAndConcavity`, §1

Back to the Car: Acceleration

Position $x(t) = t^2$ **Velocity** $v(t) = 2t$ **Acceleration** $a(t) = 2$



At $t = 3$: each slope is the next graph's height

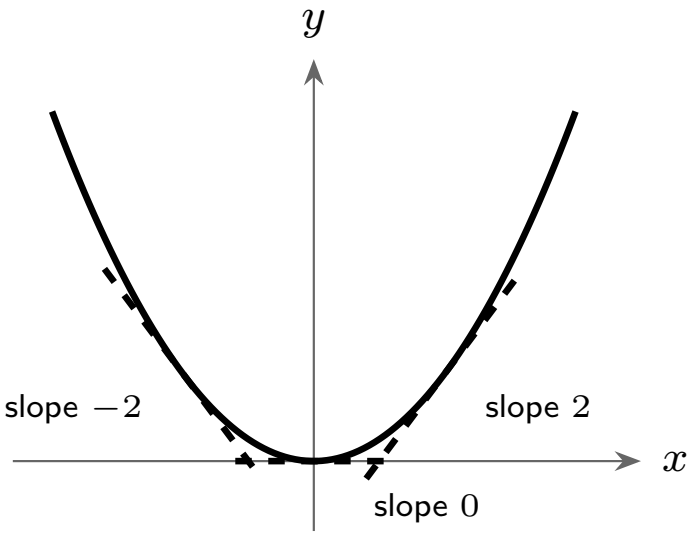
	position	**velocity**	**acceleration**
in general letters	f	f'	f''
for the car	x	$v = x'$	$a = v' = x''$
height	9 m	6 m/s	2 m/s^2
slope	6	2	0

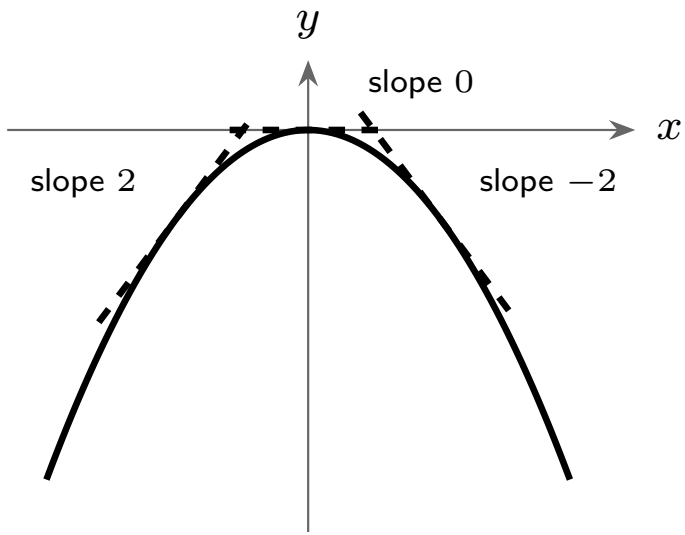
The speedometer climbs a steady 2 m/s **every second**: that is m/s^2.

Notebook `U3-3_Derivatives-3_SecondDerivativeAndConcavity`, §2

What f'' Measures: Curvature

If the slope is **increasing**, the curve curves up. If it is **decreasing**, the curve curves down.





Convex (concave up): a cup

Slopes increase, $-2 \to 0 \to 2$:
$f''(x) > 0$.

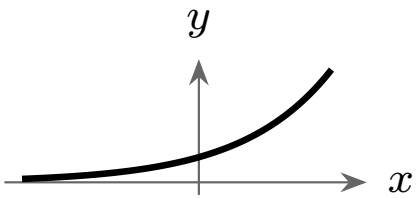
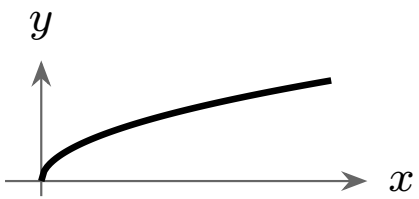
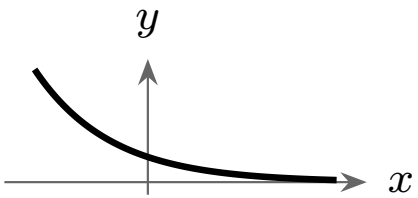
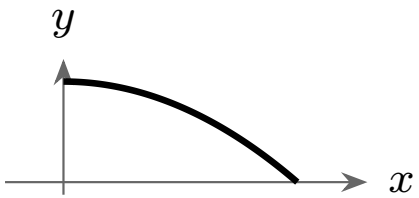
Concave (concave down): a cap

Slopes decrease, $2 \to 0 \to -2$:
$f''(x) < 0$.

Notebook `U3-3_Derivatives-3_SecondDerivativeAndConcavity`, §3

Direction and Curvature Are Separate Facts

f' says which way the curve is going. f'' says how it curves. All four combinations happen.

	$f'' > 0$: **convex**	$f'' < 0$: **concave**
$f' > 0$ **rising**		
	$y = e^x$: rising, faster and faster	$y = \sqrt{x}$: rising, more and more slowly
	an outbreak's early days	*total cases as an outbreak fades*
$f' < 0$ **falling**		
	$y = e^{-x}$: falling, more and more slowly	$y = 4 - x^2$: falling, faster and faster
	a drug clearing the blood	*a dropped ball's height*

Notebook `U3-3_Derivatives-3_SecondDerivativeAndConcavity`, §4

Where the Curvature Flips: Inflection Points

Worked: the bell curve, $h(x) = e^{-x^2/2}$

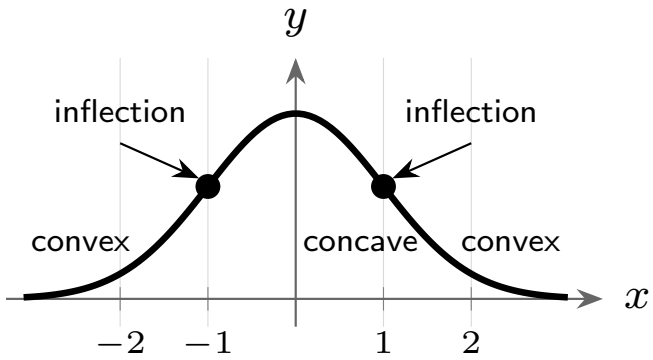
The chain rule gave $h'(x) = -x\,e^{-x^2/2}$. Now the product rule:

$$\begin{aligned} h''(x) &= (-1)\,e^{-x^2/2} + (-x)\bigl(-x\,e^{-x^2/2}\bigr) \\ &= (x^2 - 1)\,e^{-x^2/2} \end{aligned}$$

- $e^{-x^2/2} > 0$, so $x^2 - 1$ sets the sign.
- $-1 < x < 1$: $h'' < 0$, concave.
- $x < -1$ or $x > 1$: $h'' > 0$, convex.

Definition: inflection point

Where f'' **changes sign**: the curve switches between convex and concave.



What it says

The bell is steepest at $x = \pm 1$, one standard deviation from the center, because that is where the slope stops getting steeper: the curvature flips there.

Notebook `U3-3_Derivatives-3_SecondDerivativeAndConcavity`, §3

Exercise: Curvature

Try it

1. For $f(x) = 2x^3 - 6x^2 + 1$, find $f''(x)$. Where is the curve convex, where is it concave, and where is the inflection point?
2. Find $f'(x)$ and $f''(x)$ for $f(x) = \ln x$, $x > 0$. What do their signs say about the curve?
3. $C(t)$ is an outbreak's total case count after t days. On day 30, $C'(30) = 200$ cases per day and $C''(30) = -15$ cases per day per day. In plain words, what is happening?

Exercise: Solution

Answers

1. $f'(x) = 6x^2 - 12x$, so $f''(x) = 12x - 12 = 12(x - 1)$.
   - $x > 1$: $f'' > 0$, **convex**. $x < 1$: $f'' < 0$, **concave**.
   - The sign flips at $x = 1$, and $f(1) = 2 - 6 + 1 = -3$: the inflection point is $(1, -3)$.

2. $f'(x) = \dfrac{1}{x} = x^{-1}$, so by the power rule $f''(x) = -x^{-2} = -\dfrac{1}{x^2}$.

   For every $x > 0$, $f' > 0$ and $f'' < 0$: $\ln x$ is always **rising, more and more slowly**.

3. $C' > 0$: the total is still rising, by about 200 new cases a day.

   $C'' < 0$: that daily count is falling, by about 15 a day, so expect about 185 tomorrow.

   The outbreak is still growing, but it is **slowing down**.

Partial Derivatives

Functions of Two Variables, from Unit 1-3

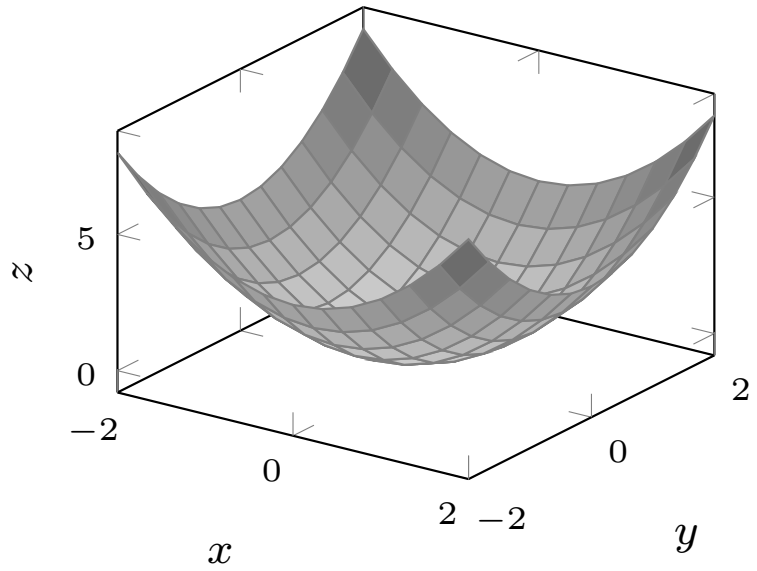
Two inputs, one output

$f(x, y)$ takes a **pair** of numbers and returns **one**. Its graph is a **surface**: each point (x, y) on the floor gets a height $z = f(x, y)$.

Examples

body mass index	$B(w, h) = w/h^2$
drug concentration	$C(D, V) = D/V$
a bowl	$f(x, y) = x^2 + y^2$
a model's loss	$L(m, b)$

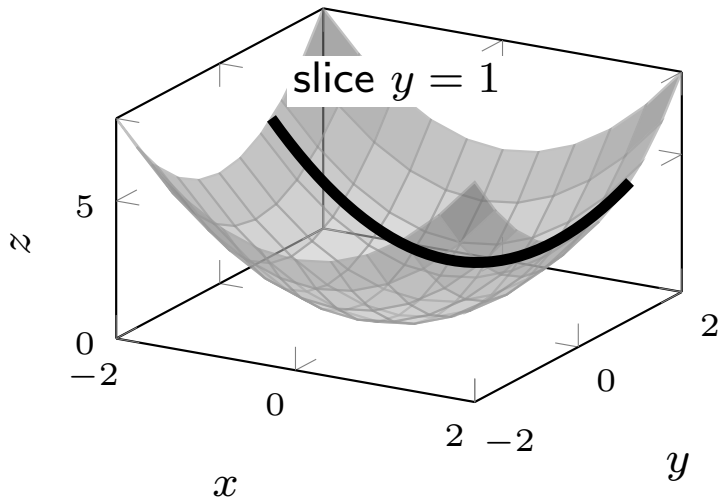
Bowl: $z = x^2 + y^2$

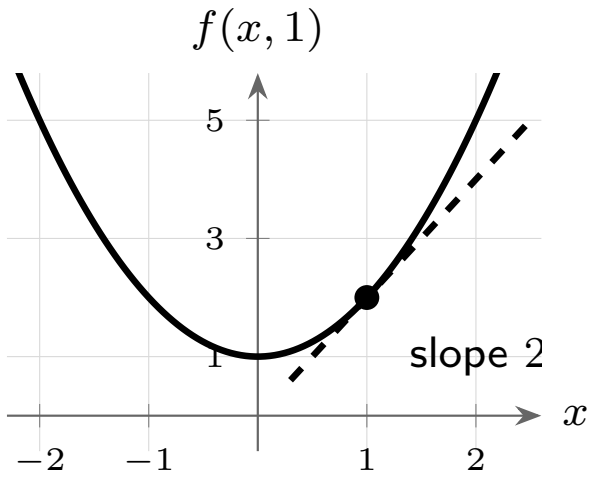


One lowest point, at $(0, 0)$.

Notebook `U3-6_Multivariable-1_Surface3D`, §1 and §2

Freeze One Variable





Dashed: the tangent at $x = 1$.

The idea, for $f(x, y) = x^2 + y^2$

- Hold $y = 1$ fixed: $f(x, 1) = x^2 + 1$ is a function of x **alone**, the slice of the bowl.
- Differentiate it with the rules from earlier: slope $2x$, so slope 2 at $x = 1$.
- That is the **partial derivative of** f **with respect to** x at $(1, 1)$.

Notebook `U3-6_Multivariable-2_PartialDerivatives`, §1 and §3

The Definition

Partial derivatives as difference quotients

$$\frac{\partial f}{\partial x} = \lim_{h \to 0} \frac{f(x+h,\, y) - f(x,y)}{h} \qquad \text{only } x \text{ moves}$$

$$\frac{\partial f}{\partial y} = \lim_{h \to 0} \frac{f(x,\, y+h) - f(x,y)}{h} \qquad \text{only } y \text{ moves}$$

The same difference quotient as before, with the other variable frozen.

Notation, and how to compute one

- ∂ is a curly d, read "partial": $\frac{\partial f}{\partial x}$ is "partial f, partial x". Shorthand: f_x and f_y.
- The curly d is a warning: other variables exist and are being held fixed.
- **In practice**: treat every other variable as a **constant**, then use the rules from earlier in this lecture.

What a Partial Means: BMI

$$B(w, h) = \frac{w}{h^2} = w\,h^{-2}$$

- Freeze h: B is w times the constant h^{-2}, so $\dfrac{\partial B}{\partial w} = \dfrac{1}{h^2}$.
- Freeze w: power rule on h^{-2}, so $\dfrac{\partial B}{\partial h} = w \cdot (-2)h^{-3} = -\dfrac{2w}{h^3}$.

For a patient with $w = 70$ kg and $h = 1.75$ m

- $\dfrac{\partial B}{\partial w} = \dfrac{1}{1.75^2} \approx 0.33$ BMI points per kg: one more kg raises BMI by about a **third** of a point.
- $\dfrac{\partial B}{\partial h} = -\dfrac{2 \cdot 70}{1.75^3} \approx -26$ BMI points per m, about -0.26 per cm: one more cm of height, at the same weight, lowers BMI by about a **quarter** of a point.

Units: a partial is in output units **per** unit of the input that moved.

Exercise: Partial Derivatives

Try it

1. For $f(x, y) = x^3 + 4xy - y^2$, find $\dfrac{\partial f}{\partial x}$ and $\dfrac{\partial f}{\partial y}$, then evaluate both at $(1, 2)$.
2. A dose D (mg) spreads through a volume V (L) of body fluid, giving the concentration $C(D, V) = \dfrac{D}{V}$.
   - Find $\dfrac{\partial C}{\partial D}$ and $\dfrac{\partial C}{\partial V}$.
   - Evaluate both at $D = 500$ and $V = 50$, with units, and say what each one means.

Exercise: Solution

Answers

1. $\dfrac{\partial f}{\partial x} = 3x^2 + 4y$ (freeze y; $4xy$ is the constant $4y$ times x), $\dfrac{\partial f}{\partial y} = 4x - 2y$ (freeze x).
   - At $(1, 2)$: $\frac{\partial f}{\partial x} = 3 + 8 = 11$ and $\frac{\partial f}{\partial y} = 4 - 4 = 0$.
   - Meaning: stepping in x, the surface climbs steeply (slope 11); stepping in y, it is momentarily **flat**.
2. Write $C = D \cdot V^{-1}$. Here $C = 500/50 = 10$ mg/L.
   - $\dfrac{\partial C}{\partial D} = \dfrac{1}{V} = \dfrac{1}{50} = 0.02$ (mg/L) per mg: each extra mg of dose raises the concentration by 0.02 mg/L.
   - $\dfrac{\partial C}{\partial V} = -\dfrac{D}{V^2} = -\dfrac{500}{2500} = -0.2$ (mg/L) per L: the same dose in a patient with 1 L more fluid gives about 0.2 mg/L **less**.

The Gradient

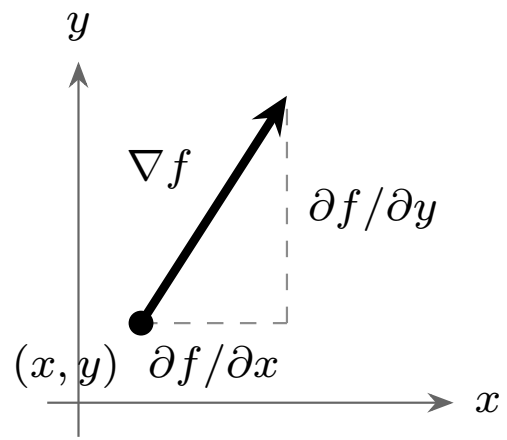
The Gradient Vector

Definition

The **gradient** of $f(x, y)$ stacks its two partials into a column vector:

$$\nabla f(x, y) = \begin{bmatrix} \partial f/\partial x \\ \partial f/\partial y \end{bmatrix}$$

- ∇ is read "del".
- It lives on the **floor**, the xy-plane of inputs: it says which way to step.



The gradient at one point.
Its two legs are the two partials.

Vector field

A **vector field** attaches a vector, with a **direction** and a **length**, to every point (x, y). We draw one as arrows on a grid of points.
The gradient is a vector field: each point gets its own arrow $\nabla f(x, y)$.

With n inputs

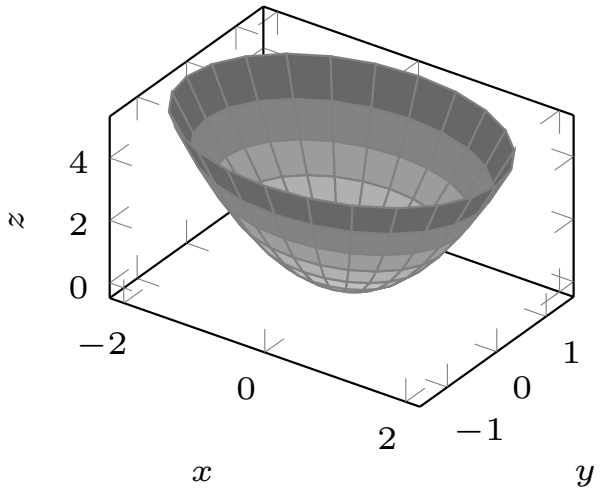
One partial per input:

$$\nabla f = \begin{bmatrix} \partial f/\partial x_1 \\ \vdots \\ \partial f/\partial x_n \end{bmatrix}$$

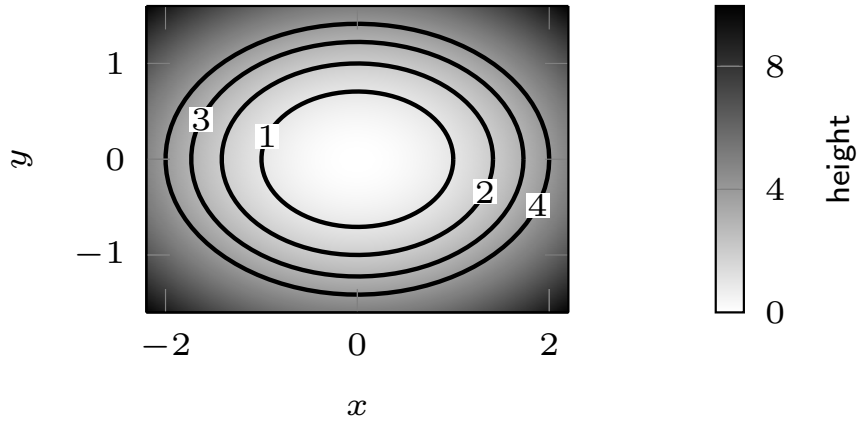
A loss with n weights has an n-entry gradient.

Four Views of $f(x, y) = x^2 + 2y^2$

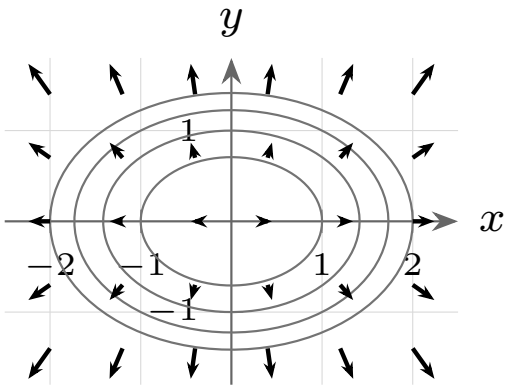
1. Surface: height $z = f(x, y)$



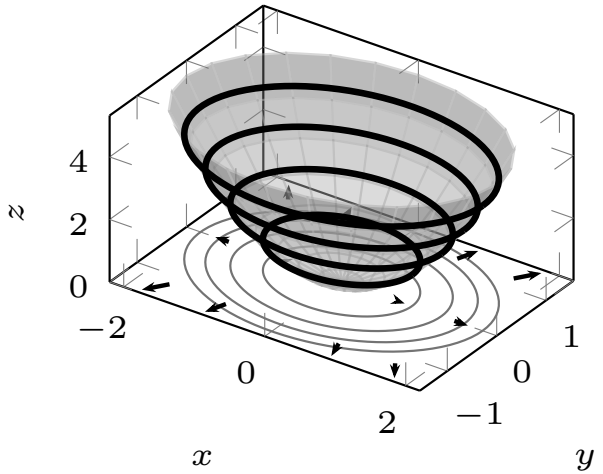
2. Heat map: shade is height



3. Gradient field: $\nabla f = \begin{bmatrix} 2x \\ 4y \end{bmatrix}$



4. Together: the floor holds views 2 and 3



Computing Gradients

$f(x, y) = x^2 + 2y^2$

$$\partial f/\partial x = 2x + 0 = 2x$$
$$\partial f/\partial y = 0 + 4y = 4y$$
$$\nabla f = \begin{bmatrix} 2x \\ 4y \end{bmatrix}, \quad \nabla f(1,2) = \begin{bmatrix} 2(1) \\ 4(2) \end{bmatrix} = \begin{bmatrix} 2 \\ 8 \end{bmatrix}$$

1. freeze y: $2y^2$ is a constant
2. freeze x: x^2 is a constant
3. stack, then plug in $(1, 2)$

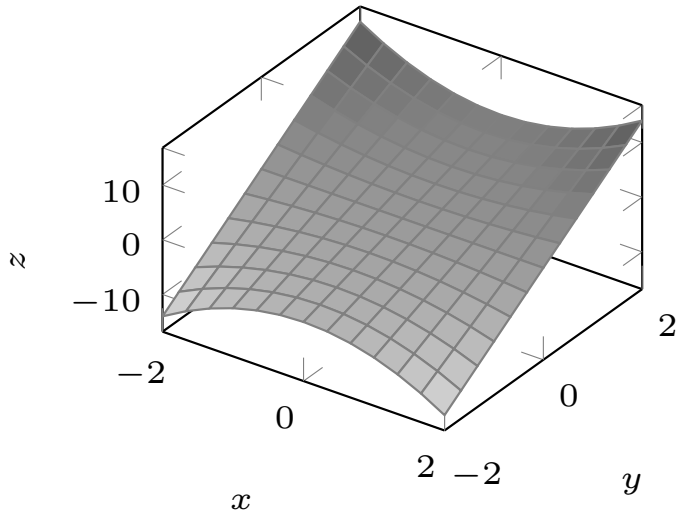
$g(x, y) = x^2 y + 3y$

$$\partial g/\partial x = 2x \cdot y + 0 = 2xy$$
$$\partial g/\partial y = x^2 \cdot 1 + 3 = x^2 + 3$$
$$\nabla g = \begin{bmatrix} 2xy \\ x^2 + 3 \end{bmatrix}, \quad \nabla g(1,2) = \begin{bmatrix} 2(1)(2) \\ 1^2 + 3 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \end{bmatrix}$$

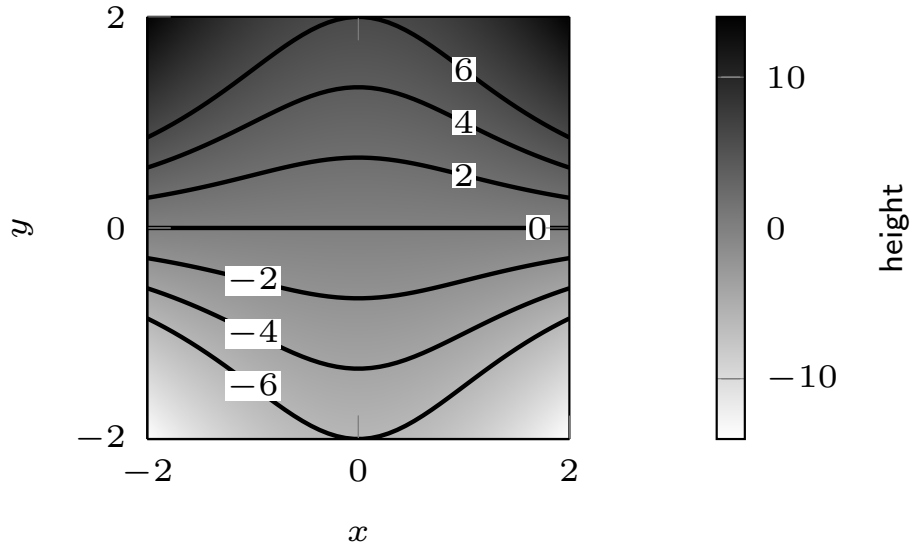
1. freeze y: $3y$ is a constant
2. freeze x: x^2 is a constant
3. stack, then plug in $(1, 2)$

Four Views of $g(x, y) = x^2y + 3y$

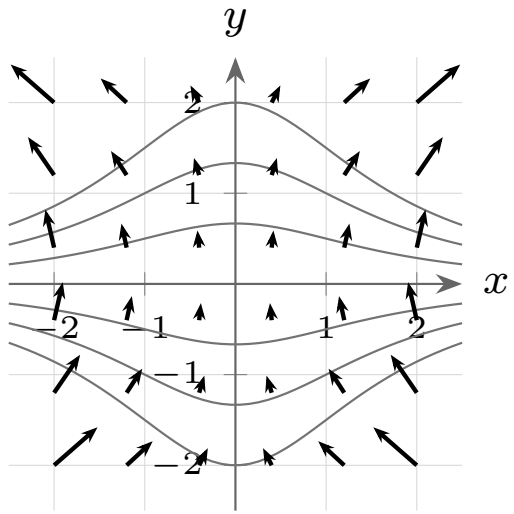
1. Surface: height $z = g(x, y)$



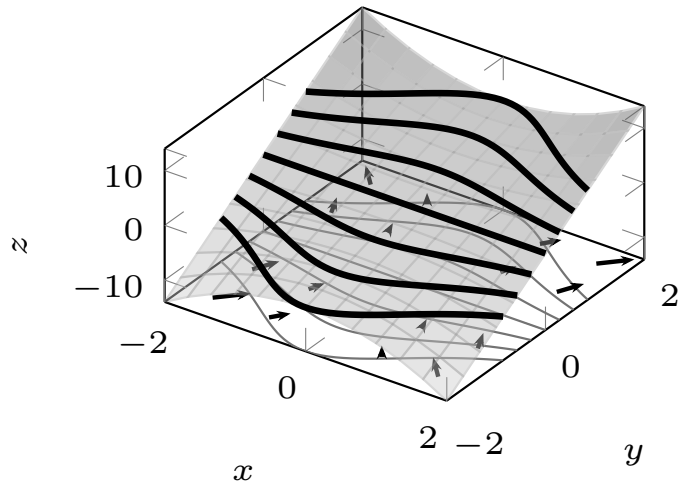
2. Heat map: shade is height



3. Gradient field: $\nabla g = \begin{bmatrix} 2xy \\ x^2+3 \end{bmatrix}$



4. Together: the floor holds views 2 and 3



The Gradient of Body Mass Index

$B(w,h) = \dfrac{w}{h^2} = w\,h^{-2}$, weight w in kg, height h in m

$$\frac{\partial B}{\partial w} = h^{-2} = \frac{1}{h^2}$$ freeze h: w times a constant

$$\frac{\partial B}{\partial h} = w \cdot (-2)h^{-3} = -\frac{2w}{h^3}$$ freeze w: power rule on h^{-2}

$$\nabla B = \begin{bmatrix} 1/h^2 \\ -2w/h^3 \end{bmatrix}$$ stack the two partials

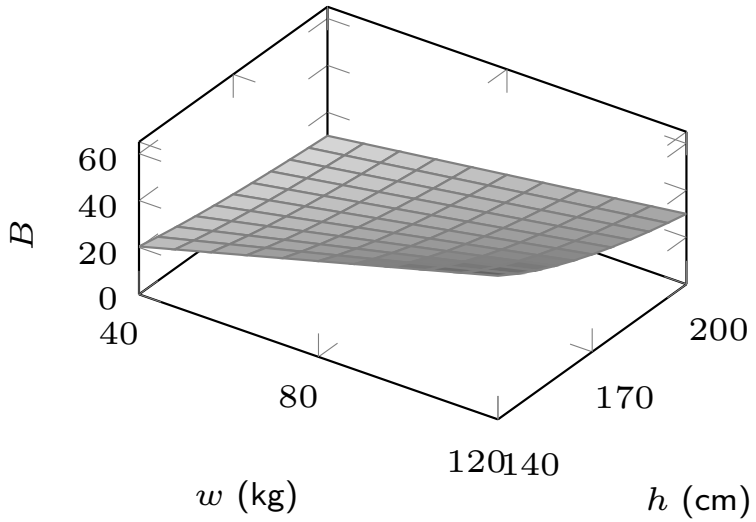
For a patient with $w = 70$ kg and $h = 1.75$ m

$$\nabla B(70, 1.75) = \begin{bmatrix} 1/1.75^2 \\ -2(70)/1.75^3 \end{bmatrix} \approx \begin{bmatrix} 0.33 \\ -26 \end{bmatrix}$$

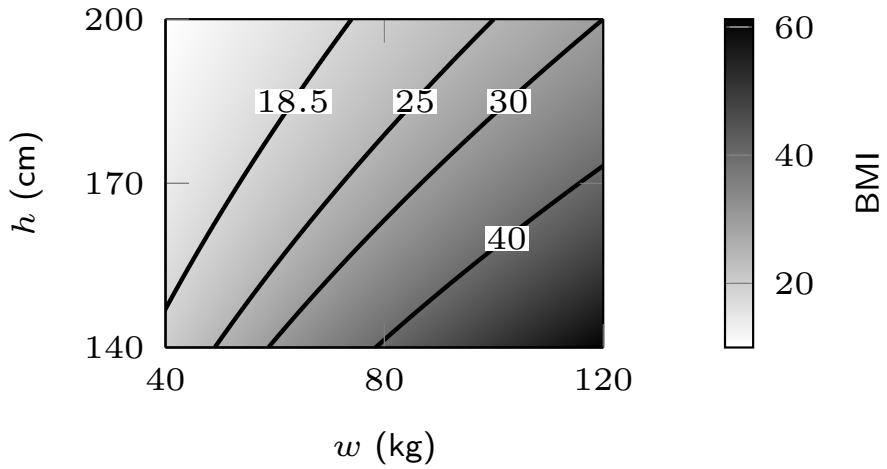
- Each entry is in BMI points per unit of its own input: 0.33 per kg, -26 per m (-0.26 per cm).
- The signs say which way BMI climbs: **up** with weight, **down** with height.

Four Views of BMI, h in cm: $B = 10000\, w/h^2$

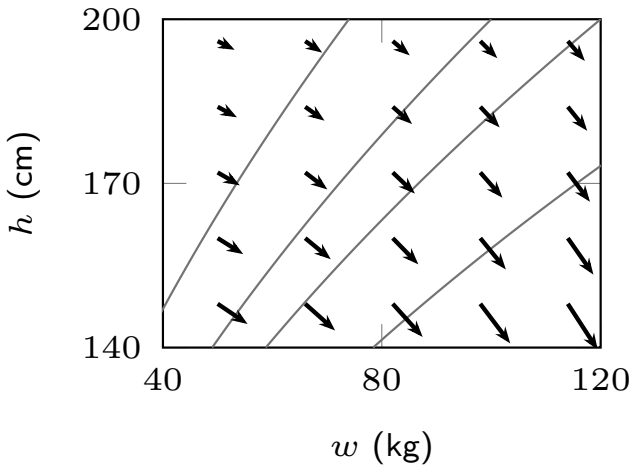
1. Surface: height $B(w, h)$



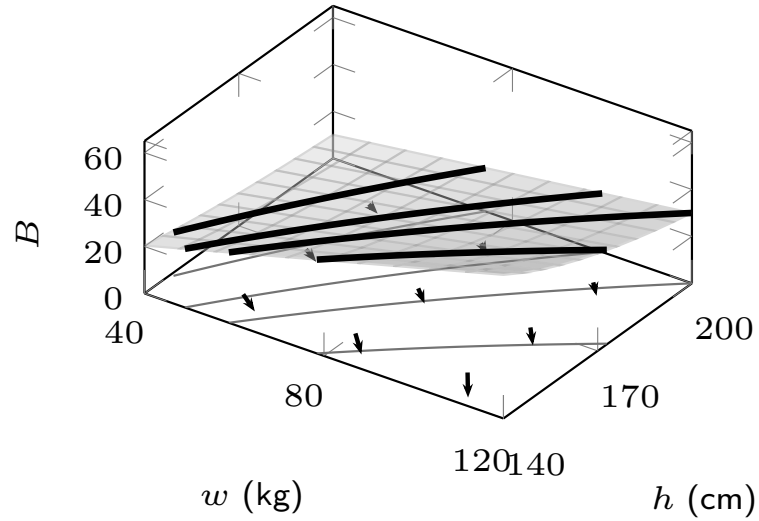
2. Heat map: shade is BMI



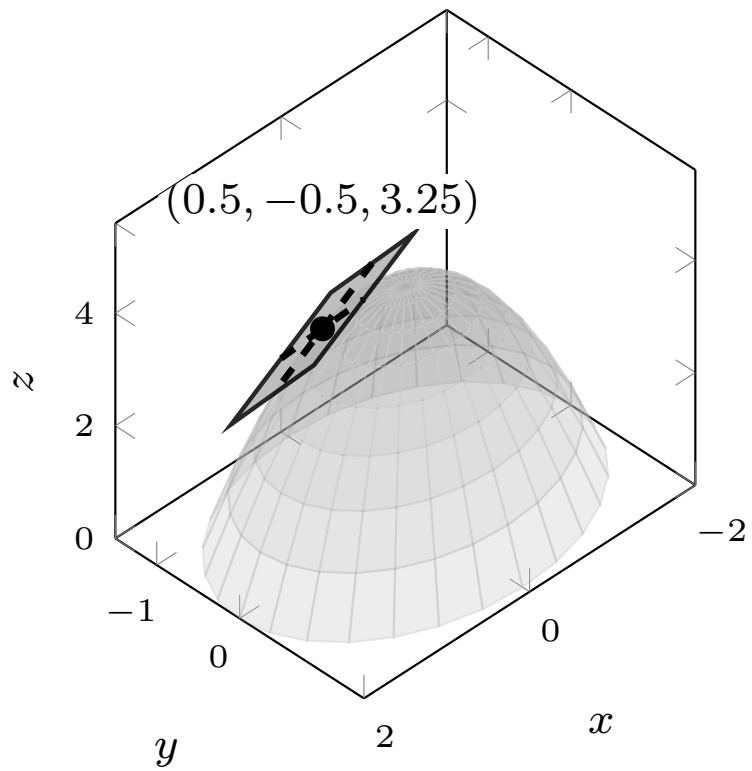
3. Gradient field of B



4. Together: the floor holds views 2 and 3



The Tangent Plane



The hill $z = 4 - x^2 - 2y^2$ (the bowl, flipped) and its tangent plane at $(0.5, -0.5)$. Dashed: the two slice tangents.

Zoom in: the surface looks flat

At (a, b) the surface touches its **tangent plane**, built from the partials at (a, b):

$$z = f(a,b) + \frac{\partial f}{\partial x}(x - a) + \frac{\partial f}{\partial y}(y - b)$$

- Freeze $y = b$: a line of slope $\partial f/\partial x$.
- Freeze $x = a$: a line of slope $\partial f/\partial y$.
- The plane holds **both** lines.

$f = 4 - x^2 - 2y^2$ at $(0.5, -0.5)$

$f = 3.25$, $\partial f/\partial x = -2x = -1$, $\partial f/\partial y = -4y = 2$:

$$\begin{aligned} z &= 3.25 - (x - 0.5) + 2(y + 0.5) \\ &= 4.75 - x + 2y \end{aligned}$$

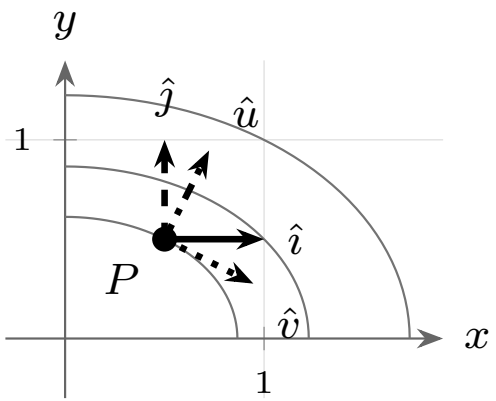
The Directional Derivative

Definition

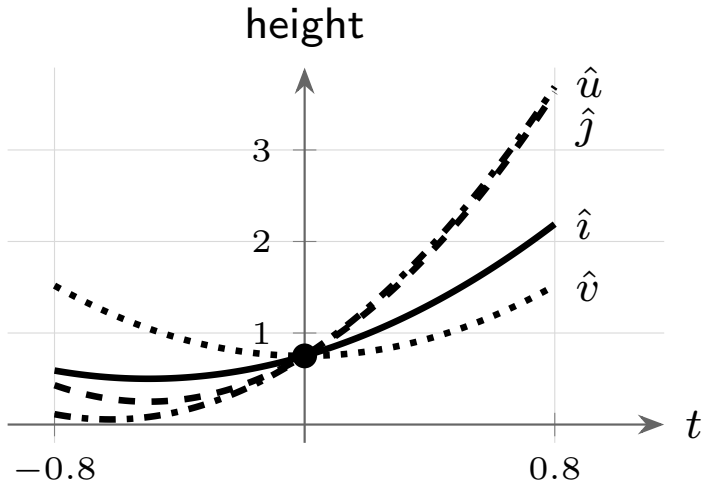
Walk from (x, y) along a unit vector $\hat{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$. The surface's slope that way is the **directional derivative**

$$D_{\hat{u}} f = \hat{u} \cdot \nabla f = u_1 \frac{\partial f}{\partial x} + u_2 \frac{\partial f}{\partial y}$$

Along the axes it is the partials: $\hat{\imath} \cdot \nabla f = \partial f / \partial x, \quad \hat{\jmath} \cdot \nabla f = \partial f / \partial y.$



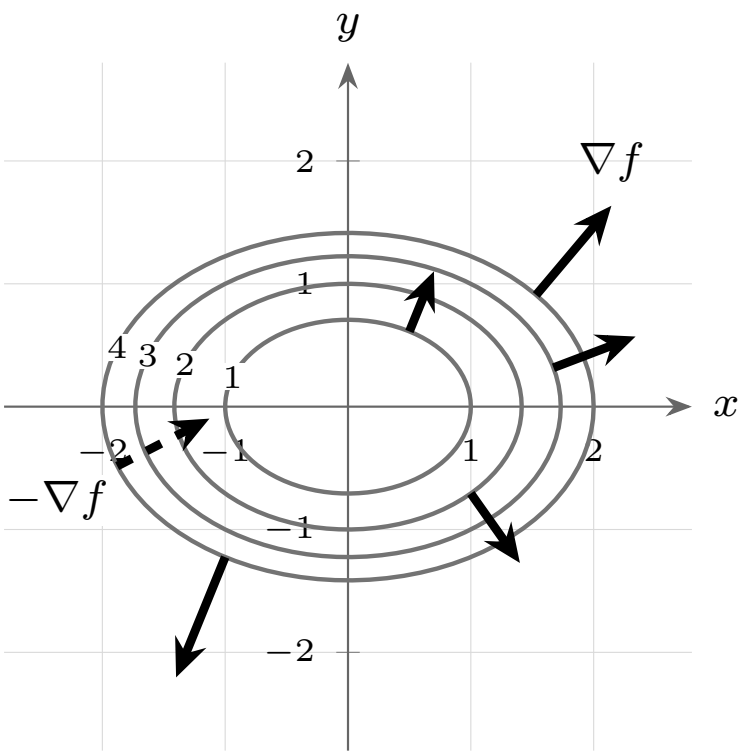
Contours of $f = x^2 + 2y^2$ at $0.75, 1.5, 3$.



Height after walking t units from P.

At P, $\nabla f = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$. Slopes: 1 ($\hat{\imath}$), 2 ($\hat{\jmath}$), $\sqrt{5} \approx 2.24$ ($\hat{u}$, along ∇f), 0 ($\hat{v}$, on the contour).

The Gradient Points Uphill



Contours of $f = x^2 + 2y^2$ at heights $1, 2, 3, 4$.
Arrows drawn at 0.2 times their length.

Why: the dot product

With θ the angle from ∇f to $\hat{u}$ (Unit 2-2):

$$D_{\hat{u}} f = \hat{u} \cdot \nabla f = \|\nabla f\| \cos\theta$$

θ	$D_{\hat{u}} f$	**walking**
$0°$	$+\|\nabla f\|$	along ∇f: steepest up
$90°$	0	along the contour: level
$180°$	$-\|\nabla f\|$	along $-\nabla f$: steepest down

- **Length** $\|\nabla f\|$: the steepest slope. Longer where contours crowd.
- Each arrow crosses its contour at a **right angle**.
- At a flat point (bottom, top, saddle), $\nabla f = \bar{0}$.

Notebook `U3-6_Multivariable-3_GradientField`, §1 to §4

Exercise: The Gradient

Try it

Let $f(x, y) = x^2 + xy$.

1. Find $\nabla f(x, y)$.
2. Evaluate it at $(4, -5)$.
3. At that point, how steep is the fastest climb? Which direction is the fastest drop?
4. Show that walking along $\hat{u} = \begin{bmatrix} 4/5 \\ -3/5 \end{bmatrix}$ has slope 0 there. What does that direction mean?

Exercise: Solution

Answers

1. In x, the term xy is the constant y times x. In y, the term x^2 is a constant and xy is x times y. So
$$\nabla f = \begin{bmatrix} 2x + y \\ x \end{bmatrix}.$$

2. $\nabla f(4, -5) = \begin{bmatrix} 8 - 5 \\ 4 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$.

3. The fastest climb is the gradient's length: $\sqrt{3^2 + 4^2} = \sqrt{25} = 5$. The fastest drop is along $-\nabla f = \left[\begin{smallmatrix} -3 \\ -4 \end{smallmatrix}\right]$, at slope -5.

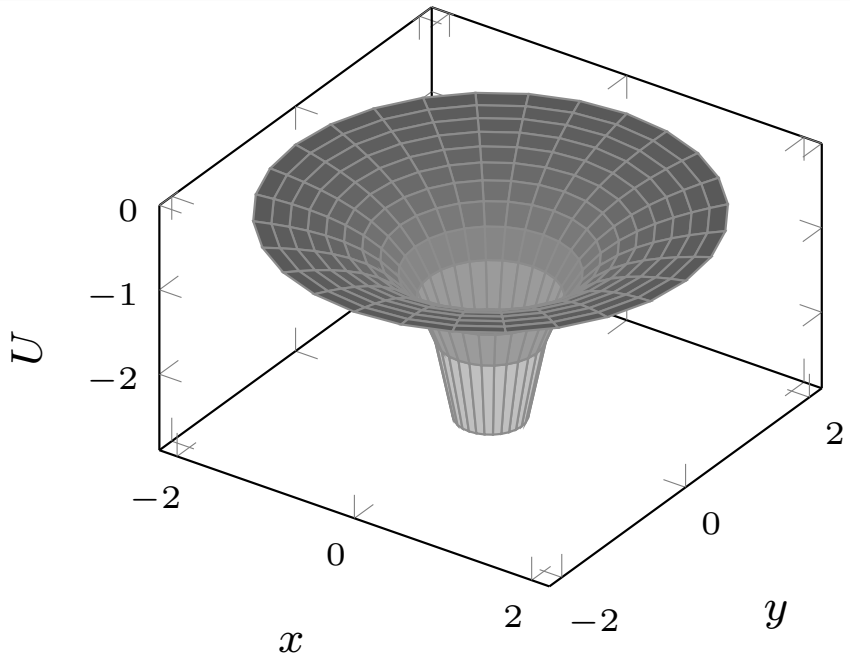
4. $\hat{u} \cdot \nabla f = \frac{4}{5}(3) + \left(-\frac{3}{5}\right)(4) = \frac{12}{5} - \frac{12}{5} = 0$.

   $\hat{u}$ is at a right angle to the gradient: it runs **along the contour**, on level ground.

Applications: Gravity's Potential Energy

Force points downhill on potential energy

Gravity stores energy in position: the **potential energy** $U(x, y)$, and the force is $\bar{F} = -\nabla U$. Near the ground $U = mgh$: on a hill of height $h(x, y)$, a ball is pushed along $-\nabla h$, straight downhill.



$U = -1/r$ around a planet at the origin (units with $GMm = 1$).

A planet of mass M

$$U = -\frac{GMm}{r}, \qquad r = \sqrt{x^2 + y^2}$$

- r: the distance to the planet.
- Close in, U is very negative: a **well**. Far out, it rises toward 0.
- A mass is pushed downhill on U: **into** the well.

Gravity: the Force Is $-\nabla U$

The gradient of $U = -GMm/r$

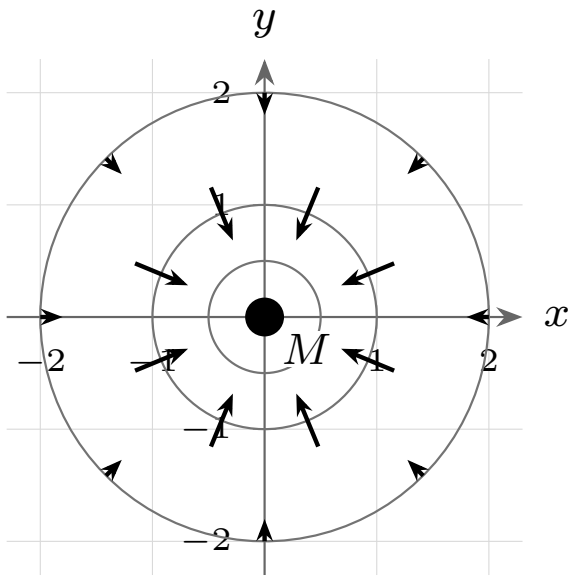
$r = \sqrt{x^2 + y^2}$ is the distance to the planet.

$$\frac{\partial r}{\partial x} = \frac{2x}{2\sqrt{x^2+y^2}} = \frac{x}{r} \quad \text{chain rule}$$

$$\frac{\partial U}{\partial x} = \frac{GMm}{r^2} \cdot \frac{x}{r} \quad \text{chain rule again}$$

$$\bar{F} = -\nabla U = -\frac{GMm}{r^3}\begin{bmatrix} x \\ y \end{bmatrix} \quad \text{same for } y$$

$\bar{F}$ points **at the planet**, with strength GMm/r^2: the **inverse square law**.



Circles: contours of U.
Arrows: the force, longer close in.

Same rule: $\bar{E} = -\nabla V$; heat flows along $-\nabla T$; drugs diffuse along $-\nabla c$.

Review: Limits and Derivatives

What we covered: from secant to tangent

1. The **average rate of change** from x_1 to x_2 is $\frac{f(x_2)-f(x_1)}{x_2-x_1}$: the slope of a **secant line**. Its units are output per input.
2. The **difference quotient** $\frac{f(x+h)-f(x)}{h}$ is the same slope, from x to $x+h$.
3. Setting $h=0$ gives $0/0$. Instead take a **limit**: let h get as close to 0 as we like, without reaching it. Simplify first, then let h shrink.
4. The **derivative** $f'(x)=\lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$ is the slope of the **tangent line**: the **instantaneous** rate of change. Also written $\frac{df}{dx}$ and $\frac{d}{dx}f(x)$.
5. The sign of f' says rising (>0), falling (<0), or flat $(=0)$.

Review: Limits and Derivatives

What we covered: the rules

6. From the definition: $c \to 0$, $x \to 1$, $mx + b \to m$, $x^2 \to 2x$, $\sin x \to \cos x$, $\frac{1}{x} \to -\frac{1}{x^2}$.
7. **Power rule**: $x^n \to n\,x^{n-1}$. Rewrite roots and fractions as powers first.
8. Constants **ride along**; sums and differences **split**.
9. **Product rule**: $(fg)' = f'g + fg'$. **Quotient rule**: $\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$.
10. The table: $e^x \to e^x$, $\ln x \to \frac{1}{x}$, $a^x \to a^x \ln a$, $\log_a x \to \frac{1}{x \ln a}$, $\sin x \to \cos x$, $\cos x \to -\sin x$.
11. **Chain rule**: for a composition $h(x) = f(g(x))$, $h'(x) = f'(g(x)) \cdot g'(x)$: the outer's derivative at the inner, times the inner's derivative.

Review: Limits and Derivatives

What we covered: continuity and differentiability

12. f is **continuous** at a when $f(a)$ exists, $\lim_{x \to a} f(x)$ exists, and the two are equal: no lifting the pen.
13. f is **differentiable** at x when the limit defining $f'(x)$ exists. That fails at **corners**, **cusps**, and **discontinuities**. Differentiable implies continuous, but not the reverse.
14. x^2 and $|x|$ are both continuous and both bottom out at 0, but only x^2 has a derivative there.

Review: Limits and Derivatives

What we covered: the second derivative

15. The **second derivative** f'', the derivative of f', measures **curvature**, how fast the slope changes: the car's acceleration.
16. $f'' > 0$ is **convex** (a cup), $f'' < 0$ **concave** (a cap), and an **inflection point** is where the sign flips. f' and f'' are separate facts: all four sign combinations happen.

Review: Limits and Derivatives

What we covered: two variables

17. A function $f(x, y)$ of two variables has a **surface** for a graph (Unit 1-3).
18. A **partial derivative** freezes the other variable and differentiates with the same rules: $\frac{\partial f}{\partial x}$ is the slope of the slice in the x direction, in output units per input unit.
19. The **gradient** $\nabla f = \begin{bmatrix} \partial f/\partial x \\ \partial f/\partial y \end{bmatrix}$ points steepest uphill, crosses contours at right angles, and its length is the steepness. $\nabla f = \bar{0}$ at flat points.
20. The slope in the direction $\hat{u}$ is $\hat{u} \cdot \nabla f = \|\nabla f\| \cos\theta$.

Notebooks: `U3-2_Limits-1`, `-2`; `U3-3_Derivatives-1` to `-3`; `U3-6_Multivariable-1` to `-3`. **Previous:** Unit 2-5, Determinants and Inverses. **Next:** Unit 3-2, **Optimization and Linear Regression**.